

Sibley School of Mechanical and Aerospace Engineering

MAE 3240 Heat Transfer

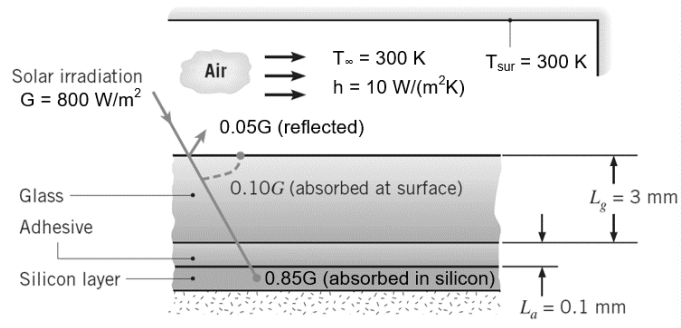
Spring 2026

Problem Set #4

Posted: Monday, Feb 23

Due: Tuesday, Mar 3, 11:59 pm

1. (40 pt) A photovoltaic (PV) panel consists of (top to bottom) a 3-mm-thick ceria-doped glass ($k_g = 1.5 \text{ W/m}\cdot\text{K}$), a 0.1-mm-thick optical grade adhesive ($k_a = 1 \text{ W/m}\cdot\text{K}$), and a very thin layer of silicon within which solar energy is converted to



electricity. The solar-to-electricity conversion efficiency within the silicon layer η decreases with increasing silicon temperature T_{Si} , as described by the expression $\eta = a - bT_{Si}$, where $a = 0.553$ and $b = 0.001 \text{ K}^{-1}$. The temperature T_{Si} is expressed in Kelvins. The backside of the silicon layer is thermally insulated. Contact thermal resistance between the adhesive and silicon layer is $R_c = 10^{-4} \text{ m}^2\cdot\text{K/W}$. There is no contact resistance between the glass and the adhesive.

In a sunny day, there is $G = 800 \text{ W/m}^2$ solar irradiation reaching the PV panel. 5% of solar irradiation is reflected from the top surface of the glass, 10% of solar irradiation is absorbed at the top surface of the glass, and 85% of solar irradiation is transmitted to and absorbed within the silicon layer. Part of the solar irradiation absorbed in the silicon is converted to thermal energy and the remainder is converted to electricity. The PV panel is exposed to ambient air with temperature of $T_\infty = 300 \text{ K}$ and convective heat transfer coefficient of $h = 10 \text{ W/m}^2\cdot\text{K}$. The surrounding temperature is $T_{sur} = T_\infty = 300 \text{ K}$ and the effective radiative heat transfer coefficient is $h_{rad} = 5 \text{ W/m}^2\cdot\text{K}$. Hint: you do not need to use the Stefan-Boltzmann law to calculate the radiative heat loss.

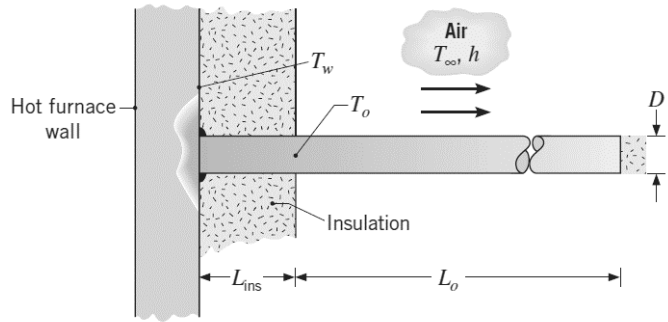
- (a) Show the thermal resistance network that can describe heat transfer across the PV panel.

Clearly label all thermal resistances, nodes for temperature, and heat fluxes entering the

temperature nodes. State key assumptions that make the thermal resistance network valid. Calculate the value of each thermal resistance in the network.

- (b) Calculate the temperatures of the top surface of the glass $T_{g,top}$ and the silicon layer T_{Si} .
- (c) Calculate the solar-to-electricity conversion efficiency η and resulting electricity power flux produced by the PV panel.

2. (30 pt) A rod of diameter $D = 10$ mm and thermal conductivity $k = 80$ W/m·K protrudes normally from a furnace wall that is at $T_w = 200$ °C and is covered by insulation of thickness $L_{ins} = 200$ mm. The rod is welded to the furnace wall and is used as a hanger for supporting



instrumentation cables. The rod has an exposed length L_o and its tip is well insulated. To avoid damaging the cables, the temperature of the rod at its exposed surface, T_o , must be maintained below a specified operating limit of $T_{max} = 100$ °C. The ambient air temperature is $T_{\infty} = 20$ °C and convective heat transfer coefficient is $h = 20$ W/m²·K.

- (a) Derive an expression for the exposed surface temperature T_o as a function of the prescribed thermal geometrical parameters.
 - (b) If exposed length L_o is 200 mm, calculate the corresponding exposed surface temperature T_o . Does the exposed length meet the specified operating limit?
 - (c) Calculate the minimum exposed length L_o that ensures T_o below the specified operating limit.
3. (30 pt) A square pin fin of uniform cross-section area is fabricated of copper ($k = 400$ W/m·K). The fin width is $s = 10$ mm and the fin is exposed to air flow with convective heat transfer coefficient $h = 100$ W/m²·K. It is known that the fin efficiency is $\eta_f = 0.6$. The fin has a convection tip.
- (a) Calculate the fin length L .
 - (b) Calculate the thermal resistance of the fin R_f and the fin effectiveness ε_f .

Problem 1:

Known: Photovoltaic panel with $L_g = 3\text{ mm}$ thick glass top layer, a $L_a = 0.1\text{ mm}$ thick adhesive, and very thin layer of silicon, with $\eta = a - bT_{si}$; $a = 0.553$ and $b = 0.001\text{ K}^{-1}$ as solar-to-electricity conv. efficiency. Contact therm. resis between silicon & adhesive $R_c = 10^{-4}\text{ m}^2\text{ K/W}$, none between glass & adhesive.

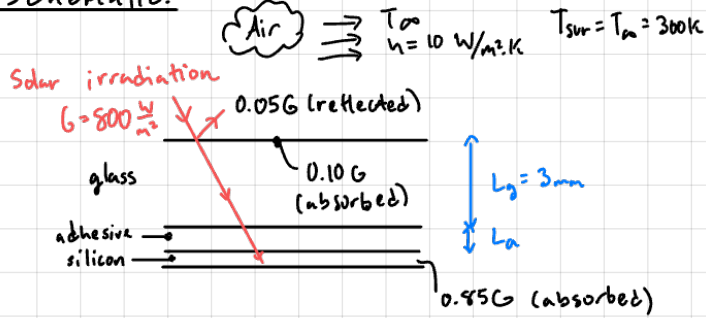
On a sunny day, $G = 800\text{ W/m}^2$ reaching panel, distributed as seen below. $T_\infty = 300\text{ K}$, $h = 10\text{ W/m}^2\text{ K}$, $T_{sur} = T_\infty = 300\text{ K}$ w/ $h_{rad} = 5\text{ W/m}^2\text{ K}$

Find: (a) Show thermal resistance network of PV panel, calc. each R_{th} in network

(b) Calculate temperatures of $T_{g,top}$ and silicon T_{si} .

(c) Calculate η & resulting electricity power flux produced by the panel

Schematic:

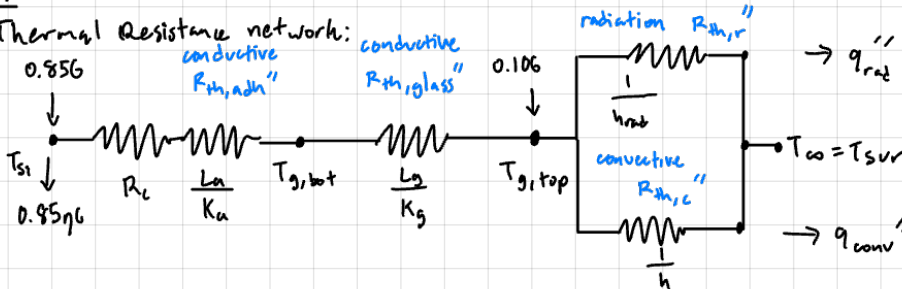


Assumptions:

- 1) Negligible thickness for silicon layer
- 2) No heat transfer @ bottom of silicon
- 3) No contact resistance btwn. glass & adhesive
- 4) Steady state conditions
- 5) 1D heat transfer
- 6) Constant Properties
- 7) Negligible temperature differences within silicon & adhesive layer

Analysis:

a) Thermal Resistance network:



See assumptions above.

$$R_{th,adh}'' = \frac{L_a}{k_a} = \frac{1 \times 10^{-3}\text{ m}}{1\text{ W/m K}} = 1 \times 10^{-3} \frac{\text{K}}{\text{W m}^2}$$

$$R_{th,glass}'' = \frac{L_g}{k_g} = \frac{0.003\text{ m}}{1.5\text{ W/m K}} = 0.002 \frac{\text{K}}{\text{W m}^2}$$

$$R_{th,r}'' = \frac{1}{h_{rad}} = \frac{1}{5\text{ W/m}^2\text{ K}} = 0.2 \frac{\text{K}}{\text{W m}^2}$$

$$R_{th,c}'' = \frac{1}{h_{conv}} = \frac{1}{10\text{ W/m}^2\text{ K}} = 0.1 \frac{\text{K}}{\text{W m}^2}$$

$$R_c = 10^{-4} \text{ m}^2 \text{ K/W}$$

$$\frac{1}{R_{th,tot}} = \frac{1}{R_{th,r}} + \frac{1}{R_{th,c}}$$

$$R_{th,tot} = \frac{1}{\frac{1}{R_{th,r}} + \frac{1}{R_{th,c}}} = 0.0667 \frac{\text{K}}{\text{W m}^2} = R_{th,tot}$$

b) S.S Energy balance on silicon layer node:

$$0.85G - 0.85\eta G + 0.10G = \frac{T_{si} - T_\infty}{R_c + R_{th,adh} + R_{th,glass} + R_{th,tot}}$$

$$0.85G(1 - (a - bT_{si})) + 0.10G = \frac{T_{si} - T_\infty}{R_{tot}}$$

$$R_{tot} = \sum R_{th} = 0.06881 \frac{\text{K}}{\text{W m}^2}$$

Solve for T_{si} :

$$0.85G - 0.85Ga + 0.85GbT_{si} + 0.1G = \frac{1}{R_{tot}}T_{si} - T_\infty \frac{1}{R_{tot}}$$

$$\Rightarrow 0.85G - 0.85Ga + 0.1G + \frac{1}{R_{tot}}T_\infty = T_{si} \left(\frac{1}{R_{tot}} - 0.85Gb \right)$$

$$\Rightarrow T_{si} = \frac{1}{\frac{1}{R_{tot}} - 0.85Gb} \left(0.85G - 0.85Ga + 0.1G + \frac{1}{R_{tot}}T_\infty \right)$$

$$T_{si} = \frac{1}{\frac{1}{0.06881 \frac{\text{K}}{\text{W m}^2}} - 0.85(800 \frac{\text{W}}{\text{m}^2})(0.001 \text{ K}^{-1})} \left(0.85 \cdot 800 \frac{\text{W}}{\text{m}^2} (1 - 0.553) + (0.1 \cdot 800 \frac{\text{W}}{\text{m}^2}) + \frac{1}{0.06881 \frac{\text{K}}{\text{W m}^2}} 300\text{ K} \right)$$

$$T_{si} = 342.44 \text{ K}$$

$q'' = q_{\text{conv}}'' + q_{\text{rad}}''$ and utilizing the thermal resistance network:

$$\frac{T_{g,\text{top}} - T_{\infty}}{R_{\text{th,tot}}} = q'', \text{ determine } q'': \frac{T_{s_i} - T_{\infty}}{R_{\text{tot}}} = q'' = \frac{342.44\text{K} - 300\text{K}}{0.06881 \frac{\text{K}}{\text{W m}^2}} = 616.77 \frac{\text{W}}{\text{m}^2}$$

$$T_{g,\text{top}} = q'' R_{\text{th,tot}} + T_{\infty} = 616.77 \frac{\text{W}}{\text{m}^2} (0.0667 \frac{\text{K}}{\text{W m}^2}) + 300\text{K} = 341.14\text{K} = T_{g,\text{top}}$$

$$(c) \quad \eta = \alpha - b T_{s_i} = 0.553 - 0.001 \frac{1}{\text{K}} (342.44\text{K}) = 0.21056 = \eta$$

The resulting electricity power flux produced by the panel is:

$$P_{\text{PV}} = 0.85 G \eta = 0.85 \cdot 800 \frac{\text{W}}{\text{m}^2} \cdot 0.21056 = 143.181 \frac{\text{W}}{\text{m}^2} = P_{\text{PV}}$$

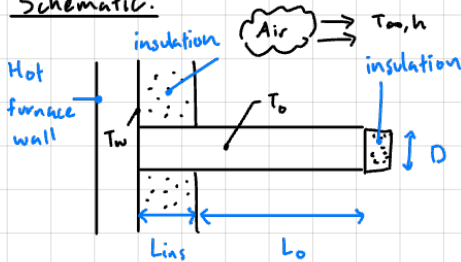
Problem 2:

Known: Rod w/ diameter $D=10\text{ mm}$, $k=80\text{ W/mK}$ protrudes from furnace wall with $T_w=200^\circ\text{C}$ and insulation $L_{ins}=200\text{ mm}$ thick. Rod is welded to wall with exposed length L_o that has insulated tip. Temp of rod @ surface T_o must be maintained below $T_{max}=100^\circ\text{C}$. $T_\infty=20^\circ\text{C}$, $h=20\text{ W/m}^2\text{K}$

Find:

- Derive expression for T_o as a function of thermal geometrical parameters
- If $L_o=200\text{ mm}$, calculate corresponding T_o . Does the exposed length meet the operating limit?
- Calculate the minimum L_o that ensures T_o below T_{max} .

Schematic:



Assumptions: 1) Steady state Conditions

2) 1D heat conduction

3) No heat loss @ insulated parts of rod (wall, L_{ins})

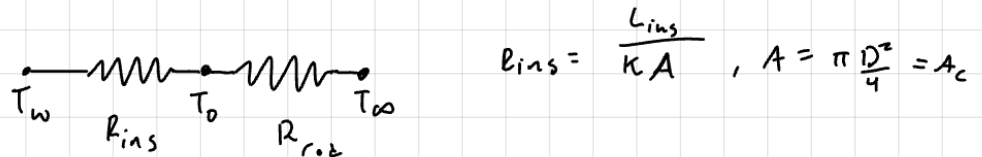
4) Negligible thermal contact resistance between rod & wall

5) Negligible radiative heat transfer

6) No heat transfer @ insulated tip

7) Fin w/ Adiabatic tip for exposed rod w/ uniform cross section

Analysis: (a) Thermal Resistance Network for Wall-Rod:



Utilizing the fin resistance formula: $R_{rod} = \frac{\theta_b}{q_f}$

From Table 3.4 for adiabatic tip: $q_f = M \tanh mL = \sqrt{hPkA_c} \theta_b \tanh mL$

Thus, $R_{rod} = \frac{1}{\sqrt{hPkA_c} \tanh(mL_o)}$ where $m = \sqrt{\frac{hP}{KA_c}}$, $P = \pi D$

From therm. network: $\frac{T_o - T_\infty}{R_{rod}} = \frac{T_w - T_\infty}{R_{ins} + R_{rod}} \Rightarrow T_o = T_\infty + \frac{R_{rod}}{R_{ins} + R_{rod}} (T_w - T_\infty)$

Expanded, $T_o = T_\infty + \left(\frac{1}{\left(\ln \pi k \frac{\pi D^3}{4} \right)^{1/2} \tanh \left(\left(\frac{4h\pi}{KD} \right)^{1/2} L_o \right)} \right) \left(\left(\frac{1}{\left(\ln \pi k \frac{\pi D^3}{4} \right)^{1/2} \tanh \left(\left(\frac{4h\pi}{KD} \right)^{1/2} L_o \right)} \right) + \frac{L_{ins}}{K \frac{\pi D^2}{4}} \right) (T_w - T_\infty)$

(b) $L_o = 0.2\text{ m}$, $\tanh(mL_o) = 0.964$, $\sqrt{hPkA_c} = 0.06283\text{ W/m}^2 = x$

$R_{ins} = 31.831\text{ K/W}$ $R_{rod} = 16.51\text{ K/W}$ Plugging in thermal + geometrical parameters

Thus $T_o = 200^\circ\text{C} + \frac{16.51\text{ K/W}}{31.831\text{ K/W} + 16.51\text{ K/W}} (200^\circ\text{C} - 20^\circ\text{C}) = 81.48^\circ\text{C} = T_o$

Since $T_o < T_{max}$, the exposed length does meet the specified operating limit.

(c) $R_{rod} = \frac{1}{x \tanh(mL_o)}$ where $x = 0.06283\text{ W/m}^2$, $m = 10\text{ m}^{-1}$

Set $T_o = 100^\circ\text{C}$ $\therefore$ solve for L_o by plugging R_{rod} in all other values are constant. using algebra solver:

$\tanh = 0.6249$

$L_{o,min} = 73.3\text{ mm}$

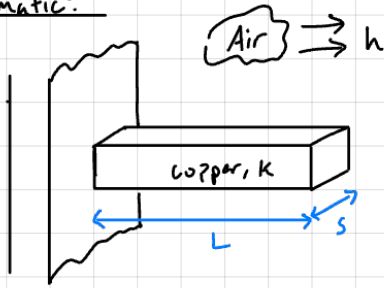
Problem 3:

Known: A square pin fin of uniform cross-section area is fabricated of copper, $K = 400 \text{ W/m}\cdot\text{K}$. Fin width $s = 10 \text{ mm}$ and fin exposed to air flow w/ $h = 100 \text{ W/m}^2\cdot\text{K}$, $\eta_f = 0.6$ and fin has a convection tip.

Find: (a) Calculate the fin length L

(b) Calculate thermal resistance of the fin $R_{t,f}$ and fin effectiveness ϵ_f .

Schematic:



Assumptions:

- 1) Steady State Conditions
- 2) Constant Properties
- 3) Neglect Thermal Radiation
- 4) 1D heat conduction

$$P = 4s$$

$$A = s^2$$

Analysis:

$$(a) \quad m = \sqrt{\frac{hP}{KA_c}} = \sqrt{\frac{100 \frac{\text{W}}{\text{m}^2\cdot\text{K}} \cdot 4 \cdot 0.01 \text{ m}}{400 \frac{\text{W}}{\text{m}\cdot\text{K}} \cdot (0.01 \text{ m})^2}} = 10 \text{ m}^{-1}$$

$$\epsilon_f = \frac{q_f}{hA_c\theta_b}$$

$$\eta_f = \frac{q_f}{hA_f\theta_b} = \frac{\tanh h(mL_c)}{mL_c} \quad \text{thus} \quad 0.6 = \frac{\tanh(10 \text{ m}^{-1} \cdot L)}{10 \text{ m}^{-1} L_c}, \quad \text{solve for } \frac{\tanh(x)}{x} = 0.6$$

$$\Rightarrow x = 1.51222 \quad \therefore x = mL \Rightarrow L_c = \frac{1.51222}{10 \text{ m}^{-1}} = 0.1512 \text{ m} = L_c$$

Corrected fin length for convective tip: $L_c = L + \frac{t}{2}$

$$\text{thus } L = L_c - 0.01 \text{ m} / 2 = \boxed{0.1462 \text{ m} = L}$$

$$(b) \quad \epsilon_f = \frac{M \tanh(mL_c)}{hA\theta_b} = \frac{\sqrt{hPK} \tanh(mL_c)}{hA}$$

$$R_{t,f} = \frac{1}{hA_f\eta_f}$$

$$\epsilon_f = \frac{\sqrt{h4sK}}{hs^2} \tanh(mL_c)$$

$$\epsilon_f = \frac{\sqrt{100 \cdot 4 \cdot 0.01 \text{ m} \cdot 400}}{100 \cdot (0.01 \text{ m})^2} \tanh(10 \text{ m}^{-1} \cdot 0.1512 \text{ m})$$

$$\boxed{\epsilon_f = 3629.16}$$

$$A_f = 4L_s + s^2$$

$$A_f = 4(0.1462 \text{ m})(0.01 \text{ m}) + (0.01 \text{ m})^2$$

$$A_f = 0.005948 \text{ m}^2$$

$$R_{t,f} = \frac{1}{100 \frac{\text{W}}{\text{m}^2\cdot\text{K}} \cdot 0.005948 \text{ m}^2 \cdot 0.6}$$

$$\boxed{R_{t,f} = 2.8021 \frac{\text{K}}{\text{W}}}$$